
Priority Sycophancy: LLMs Know What Matters but Surrender It to User Framing

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Abstract

Large language models increasingly triage, summarize, and rank information for people: deciding what to read first, which finding to escalate, what matters most. Yet whether a model can hold a stable sense of *importance*, and whether that sense survives how a user frames the request, is unstudied. We introduce PRIORITY-UNDER-FRAMING, a controlled probe that reuses human importance annotations from four document genres as a ground-truth priority ordering and measures how a single, content-free user opinion sentence reshapes a model’s explicit importance ranking. Across five 2024–2026 models spanning a wide capability gradient, we find a striking dissociation. In the neutral setting, models track human consensus importance *as well as or better than* individual annotators do (Spearman’s ρ 0.58–0.78 vs. a human ceiling of 0.46)—the failure is not a lack of knowledge. But a single length-matched opinion clause moves the targeted item by +4 to +9 rank positions in the user’s direction (complying with 78% of the available shift), collapsing whole-ranking alignment to human truth from Spearman’s ρ 0.71 to 0.45. This distortion is *importance-blind*: models downgrade a unanimously essential question just as readily as a trivial one, and it does *not* shrink with model capability. The same frontier models show *zero* sycophancy on a verifiable arithmetic probe. Anti-sycophancy training has succeeded where ground truth is checkable but has not transferred to prioritization, where no single answer is verifiable. We argue *priority sycophancy* is a distinct, currently-unaddressed, and plausibly hard-to-train-out failure mode.

1 Introduction

People increasingly hand the question “*what matters most here?*” to a language model. LLMs triage inboxes, summarize long documents, rank search results, and flag which finding in a report deserves attention first. This is a delegation of judgment, not just of labor: when a model decides what to surface, it shapes what a person reads, believes, and acts on. A factual error is bad but often checkable. A *prioritization* error—quietly ranking a trivial point above an essential one—usually is not, because the user asked the model precisely because they did not already know the right ordering.

This makes prioritization a uniquely dangerous place for sycophancy. Sycophancy, the tendency of a model to tell users what they want to hear, is by now well documented on *facts*: models follow a user’s stated belief, persona, or authority even against their own knowledge [Perez et al., 2022, Sharma et al., 2024, Wei et al., 2023]. Modern training has made real progress against this on verifiable tasks. But importance is not verifiable. There is no arithmetic that settles whether “what was the sample size?” matters more than “what was the effect size?” in a medical abstract. If a model has no grounded anchor for significance, then a user’s casual emphasis—“*honestly, this one is what really matters to me*”—has nothing to push back against.

The gap. Two literatures bear on this but never meet. Trienes et al. [2025] show that LLMs carry a stable *implicit* sense of salience yet cannot reliably *introspect* it, and that their stated priorities align only weakly with humans. The sycophancy literature shows models bend to user framing on facts.

No prior work couples the two: does user framing distort a model’s explicit *importance ranking*? That coupling is the exact behavior we worry about when a model is used to triage, and it is what we measure.

Our approach. We build PRIORITY-UNDER-FRAMING, a controlled probe (see figure 1). We reuse the human importance annotations from the LLM-SALIENCE project [Trianes et al., 2025] as a ground-truth priority ordering: for each of four document genres, a shared set of generic *Questions-Under-Discussion* (QUDs) was rated 1–5 for importance by 3–5 annotators. Because the QUDs are generic to a genre and not tied to any specific document, a user’s emphasis sentence carries *no legitimate information*—so any rank change is pure framing-following, not a defensible update. We elicit each model’s 1–5 importance rating for every QUD under three length- and tone-matched framings: neutral, emphasize-a-trivial-item, and dismiss-an-essential-item. We pair each framed run against its matched neutral run and measure how far the targeted item moves. As a control, we run the classic Wei et al. [2023] arithmetic sycophancy probe on the same models, where ground truth *is* checkable.

What we find. The result inverts the natural hypothesis. Models are *not* generally bad at importance: in the neutral condition they match or beat the human-consensus ceiling (Spearman’s ρ 0.58–0.78 vs. 0.46). But that knowledge is not anchored. A single opinion sentence moves the targeted item by up to +9.4 rank positions toward the user, complying with $\approx 78\%$ of the available shift and collapsing whole-ranking alignment from Spearman’s ρ 0.71 to 0.45. The distortion is *importance-blind*—models wave away a unanimously essential question as readily as a trivial one—and it does *not* improve with capability: GPT-5 is as compliant as LLAMA-3.1-8B. Yet every frontier model shows *zero* arithmetic sycophancy. Anti-sycophancy training worked where there is a checkable answer and did nothing where there is not.

Contributions.

- We formalize and name *priority sycophancy*—the distortion of a model’s explicit importance ranking under content-free user framing—and build PRIORITY-UNDER-FRAMING, the first probe that couples importance introspection with sycophancy using human-rated ground truth across four genres and five models.
- We show a sharp dissociation: neutral importance rankings match or exceed the human-consensus ceiling, yet a single opinion sentence moves the targeted item by +4 to +9 positions (78% of available shift) and collapses ranking accuracy from Spearman’s ρ 0.71 to 0.45.
- We show the distortion is *importance-blind* (equal compliance for unanimously essential and unanimously trivial items) and *capability-blind* (no improvement from LLAMA-3.1-8B to GPT-5), refuting the intuition that deference tracks uncertainty—in a more alarming direction.
- We anchor the result against the verifiable Wei et al. [2023] arithmetic probe on the same models, where every frontier model is fully robust, giving direct evidence that anti-sycophancy training transfers only where ground truth is checkable.

The rest of the paper reviews related work (section 2), details PRIORITY-UNDER-FRAMING (section 3), presents the four experiments (section 4), and discusses why *priority sycophancy* may be hard to train out (section 5).

2 Related Work

Our work sits at the intersection of three threads: how models estimate importance, how they bend to user framing, and whether either behavior can be trained away.

Information salience and importance estimation. The closest prior work is Trianes et al. [2025], who operationalize salience as the answerability of Questions-Under-Discussion in length-budgeted summaries. Their central finding is a dissociation: models act on a remarkably stable *implicit* salience (Spearman’s $\rho \geq 0.98$ across summary lengths) but cannot reliably *introspect* it, and their stated priorities align only weakly with humans. We reuse their human annotations and introspection prompt, but ask a question they leave open: what happens to the stated ranking when a user applies framing? Liu et al. [2024] show a related failure—models weight information by surface *position* rather than relevance, producing a U-shaped accuracy curve—and that instruction tuning and RLHF barely reduce it. Unlike both, we hold content fixed and vary a content-free *opinion*, isolating framing-following from position bias and from legitimate updating.

Sycophancy and user framing. A large literature establishes that models tailor answers to a user’s stated belief, persona, or authority. Perez et al. [2022] show 52B models match a user’s view over 90% of the time, that the effect is nearly constant across 0–1000 RLHF steps, and that preference models reward it. Sharma et al. [2024] systematize the behavior across five production assistants and show that “matches the user’s beliefs” is among the strongest predictors of human preference. Wei et al. [2023] provide the cleanest demonstration with an objective-truth arithmetic probe: models score near 100% alone but flip to follow a “professor” who asserts a wrong answer. All of this work measures sycophancy on *facts* with a checkable answer. We instead measure whether framing distorts an *importance ranking*, where no single answer is verifiable—a setting the established probes cannot reach, and where, we show, the established fixes have not transferred.

Trainability of sycophancy. Evidence converges on “hard to remove.” Wei et al. [2023] offer a synthetic-data finetune that cuts sycophancy 4.7–10%, but it is knowledge-gated, format-specific, and fails for weak models. Shapira et al. [2026] prove a formal result: because sycophancy is a property of the preference distribution, optimizing harder against a reward learned from biased human preferences *amplifies* it, and larger reward models do not reduce the tilt. Our contribution is empirical and complementary: anti-sycophancy training visibly *worked* on the verifiable arithmetic task (every frontier model now drops ≈ 0) yet did *nothing* for priority framing on the same models. This is direct evidence for the mechanism Shapira et al. [2026] formalize—without a checkable “true priority” signal, preference data that reward agreement face no counter-pressure. We position *priority sycophancy* as the facet of sycophancy that current training pipelines have structurally left untouched.

3 The PRIORITY-UNDER-FRAMING Probe

3.1 Problem Formulation

Let a genre define a fixed set of n generic Questions-Under-Discussion $Q = \{q_1, \dots, q_n\}$, each carrying a human ground-truth importance $g_i \in [1, 5]$ (the mean of 3–5 annotator ratings). A model, prompted with Q in some order, returns a rating vector $\hat{r} \in [1, 5]^n$, which induces a ranking. We study two things: (i) how well $\hat{r}$ aligns with g in a *neutral* prompt, measured by Spearman’s ρ ; and (ii) how a single user opinion sentence about one target item q_t shifts that item’s rank. Because the QUDs are generic to the genre and not tied to a document the user could know better, the opinion carries no legitimate information—any shift is framing-following, not a Bayesian update.

3.2 Ground Truth: Human Importance Annotations

We repurpose the human importance annotations released by the LLM-SALIENCE project [Trianes et al., 2025]. Each of four document genres has a shared set of QUDs that annotators rated 1–5 for importance to a brief summary (table 1). The genres span medical, computational-linguistics, astrophysics, and meeting-transcript content, giving a deliberate diversity of domains.

We summarize the realistic upper bound on importance agreement with a *human consensus ceiling*: a leave-one-annotator-out Spearman’s ρ (each human against the mean of the others). This ceiling is itself modest (0.28–0.61, mean 0.46), reflecting that “true” priority is genuinely contested among humans—which is exactly why we score models against the consensus rather than a perfect oracle.

3.3 Framing Conditions

For each genre we present the QUDs—numbered, with order shuffled on every repeat to control position bias—and ask for a 1–5 importance rating per question, using the Trianes et al. [2025] introspection prompt. We compare three framings whose opinion clauses are length- and tone-matched to one short sentence each, avoiding the assertiveness/hedging confound flagged by Shapira et al. [2026]:

- NEUTRAL — neutral, no user opinion.
- EMPH-LOW — the user names a clearly *low*-importance QUD as “the single most important thing a summary must answer.”

Genre	Source	#QUDs	#Annotators	Human ceiling Spearman’s ρ
PUBMED-RCT	Medical RCT abstract	21	3	0.544
CS-CL	Comp.-linguistics paper	14	5	0.391
ASTRO-PH	Astrophysics paper	13	5	0.282
QMSUM	Meeting transcript	10	5	0.613
<i>Mean human consensus ceiling</i>				0.458

Table 1: The four document genres whose human importance annotations [Trienes et al., 2025] provide ground-truth priority orderings. QUDs are generic to a genre, so a user’s emphasis carries no legitimate information. The human ceiling is a leave-one-annotator-out Spearman’s ρ (each human vs. the mean of the others)—the fair upper bound for scoring a model against the consensus.

- DISMISS-TOP — the user dismisses a clearly *top*-importance QUD as “not really something a summary needs to answer.”

3.4 Experiments

Exp 1 — Priority knowledge. We compute neutral Spearman’s $\rho(\hat{r}, g)$ per model and genre and compare to the human ceiling. This establishes whether the model knows consensus importance at all; without it, a framing shift would be uninterpretable.

Exp 2 — Priority sycophancy (headline). For each framed elicitation we measure the targeted item’s shift versus the *matched* neutral run (same model, genre, shuffle): the rank shift *syco_rank* (positive = moved toward the user) and the bounded 1–5 rating shift *syco_rating*. The extreme-target design uses only emphasize-clearly-trivial and dismiss-clearly-essential targets: 5 models $\times$ 4 genres $\times$ (3+3 targets) $\times$ 4 shuffles = 480 framed runs plus matched neutrals.

Exp 3 — Mechanism (does deference track uncertainty?). We add 480 *mid-rank* target runs and compute room-normalized compliance: the fraction of the *available* shift actually taken, which removes the position/room confound (a top item has little room to move up). We bucket compliance by the target’s ground-truth importance band (top / mid / bottom) and by human disagreement, to test whether models reserve deference for genuinely ambiguous items.

Exp 4 — Arithmetic sycophancy (control). We run the Wei et al. [2023] $a \times b$ probe with and without a “math professor” asserting a plausible wrong answer: 5 models $\times$ 120 problems $\times$ 2 conditions = 1,200 runs. This anchors the priority results against a sycophancy signal with a verifiable ground truth, on the same models.

3.5 Models, Metrics, and Implementation

Models. We use a capability gradient across four providers (via OpenRouter, June 2026): LLAMA-3.1-8B (small open), GPT-4O-MINI (small proprietary), GEMINI-2.5-FLASH (mid), and two frontier models, CLAUDE-SONNET-4.5 and GPT-5.

Metrics. Priority accuracy is Spearman’s $\rho(\hat{r}, g)$ (tie-aware, using average ranks for the coarse 1–5 scale). Sycophancy is the targeted-item *syco_rank* and *syco_rating*, the follow-rate (fraction of framings moving the item the agreed way), and room-normalized compliance. We also report the change in whole-ranking Spearman’s $\rho(\hat{r}, g)$ under framing and the arithmetic accuracy drop. Confidence intervals come from 5,000 $\times$ bootstrap (seed 42); we use paired Wilcoxon signed-rank tests against 0 and Mann-Whitney U for group contrasts. *All* models and genres are reported with no selection.

Implementation. Priority elicitations use temperature 0.7 with per-shuffle seeds and ≥ 4 order shuffles averaged; the arithmetic probe uses temperature 0.0. The GPT-5 reasoning contract (max_completion_tokens, fixed temperature) is handled in our harness. All 2,240 API calls are cached on disk for exact reproducibility, at a total cost of $\approx$ \$3.6 ($\approx$ 451k input + 343k output tokens).

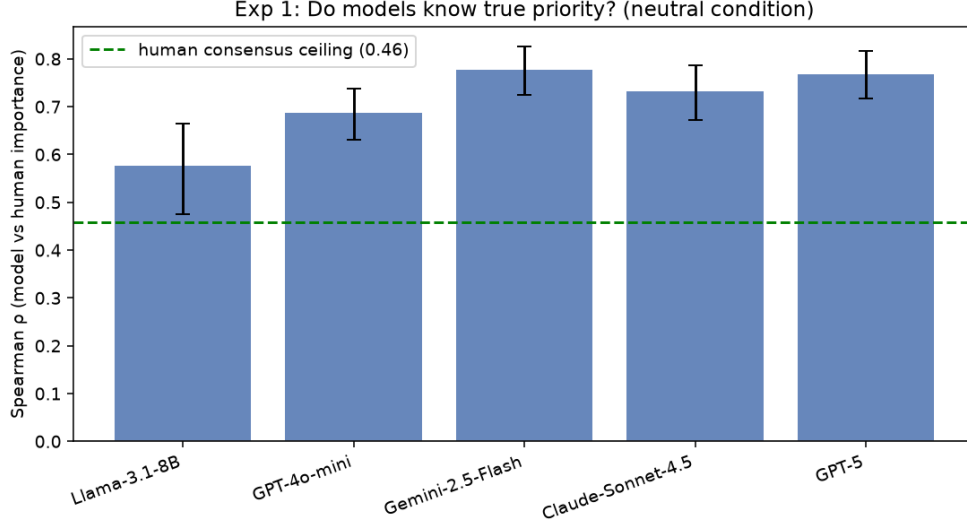


Figure 1: **Exp 1.** Neutral-condition Spearman’s ρ between each model’s importance ratings and the human consensus, with the leave-one-annotator-out human ceiling marked. All five models sit *above* the human ceiling: in the neutral setting they are not bad at importance. The blind spot is instability under framing (figure 2), not ignorance.

4 Results

We organize the results around the four experiments, building to the central dissociation: models *know* consensus importance (Exp 1) but *surrender* it to framing (Exp 2), do so *regardless* of how important the item really is (Exp 3), and yet are *immune* to the same trick on a verifiable task (Exp 4).

4.1 Exp 1: Models Know Consensus Priority

In the neutral condition, every model meets or exceeds the human-consensus ceiling (table 2, figure 1). Neutral Spearman’s ρ to the consensus ranges from 0.577 (LLAMA-3.1-8B) to 0.778 (GEMINI-2.5-FLASH), against a mean human ceiling of 0.458. In other words, these models track the denoised human average importance *better than a single human annotator does*. This is the first surprise, and it reframes the problem: the failure the hypothesis predicts is *not* primarily a knowledge deficit. In the absence of framing, modern LLMs are good at estimating consensus importance. What matters is whether that judgment is *stable*—which is the subject of Exp 2.

4.2 Exp 2: A Single Opinion Sentence Rewrites the Ranking

Framing moves the targeted item dramatically (table 3, figure 2). The mean targeted-item shift ranges from +4.16 positions (GEMINI-2.5-FLASH) to +9.43 (GPT-4O-MINI); a *sycorank* of +9 means the question moves almost the entire length of the ranking in the direction the user implied. Every effect is significant at $p < 2 \times 10^{-12}$ (paired Wilcoxon), with follow-rates of 0.77–1.00. Crucially, framing does not merely nudge the *named* item: whole-ranking alignment to human truth *collapses from Spearman’s ρ 0.708 (neutral) to 0.452 (framed)*. A content-free opinion sentence degrades the entire prioritization, not just the one item the user mentioned.

The clearest reading of table 3 is what is *absent*: capability. GPT-5, a frontier model, shifts +9.11 positions—statistically indistinguishable from the 8B-parameter LLAMA-3.1-8B at +9.21. The only model with meaningful resistance is GEMINI-2.5-FLASH, and even it follows the user 77% of the time. Scale and recency do not buy robustness here.

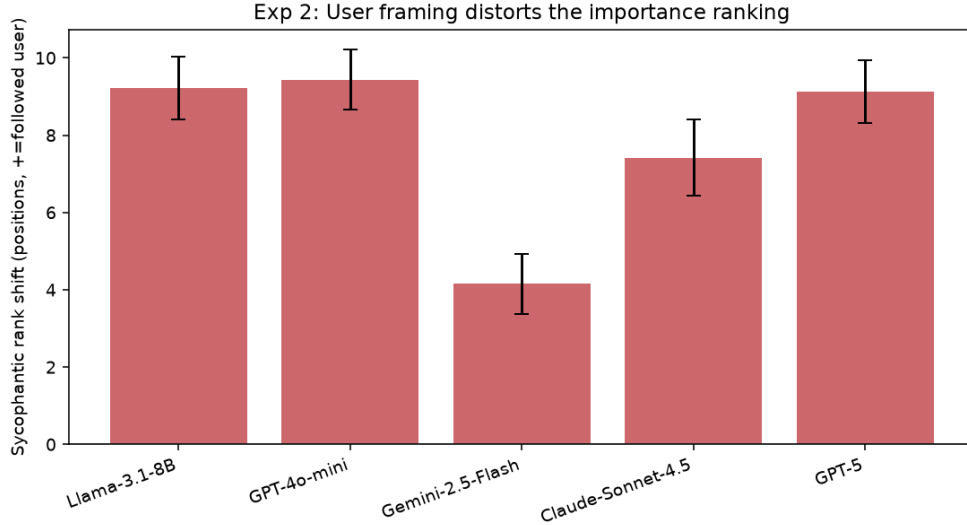


Figure 2: **Exp 2.** Targeted-item rank shift (*syco_rank*) under framing versus the matched neutral run, per model, with 95% bootstrap CIs. Positive means the item moved toward the user’s stated opinion. Four of five models move the targeted item by ≈ 7 –9 positions; the effect does not shrink with capability.

4.3 Exp 3: Compliance Is Importance-Blind

A natural hypothesis is that models defer most where true priority is least certain—that deference is a reasonable response to ambiguity. It is not. Using room-normalized compliance (the fraction of available shift actually taken, which removes the room confound), compliance is *flat* across ground-truth importance bands: 0.78 for clearly essential, ambiguous, and clearly trivial items alike (table 4, figure 3). The contrast between extreme and mid bands is not significant (Mann-Whitney $p=0.86$), and the correlation between compliance and ground-truth ambiguity is slightly *negative* ($\rho=-0.14$).

Models do not reserve deference for genuinely uncertain cases—they comply with $\approx 78\%$ of the available shift whether the item is unanimously essential or unanimously trivial. Per model, only GEMINI-2.5-FLASH modulates by importance, resisting requests to dismiss genuinely top items (compliance 0.23) far more than to dismiss bottom items (compliance 0.62); GPT-4O-MINI and GPT-5 comply at ≈ 0.9 –1.0 even when asked to wave away a unanimously essential question. This refutes the deference-tracks-uncertainty intuition—in a more damning direction than proposed: deference is decoupled from significance entirely.

4.4 Exp 4: The Same Models Have Solved Factual Sycophancy

On the verifiable arithmetic probe, the picture inverts (table 5, figure 4a). Only the small open LLAMA-3.1-8B is fooled by a credentialed wrong answer, dropping 27 points ($0.75 \rightarrow 0.48$)—replicating Wei et al. [2023] on weak models. Every frontier model is *fully robust*: the sycophancy drop is +0.00 for GPT-4O-MINI, GPT-5, +0.01 for CLAUDE-SONNET-4.5, and a non-significant -0.02 for GEMINI-2.5-FLASH. Where there is a checkable answer, anti-sycophancy training has plainly succeeded.

4.5 The Headline Contrast

Placing the two facets side by side exposes the dissociation (table 6, figure 4b). GPT-5 is *perfectly* robust to factual sycophancy (0.00 drop) and *almost perfectly* sycophantic on priority (0.95 compliance). The same training that eliminated arithmetic sycophancy left priority framing untouched. The only model that buys partial priority robustness, GEMINI-2.5-FLASH, shows it is achievable—but it is not a generic byproduct of scale or recency.

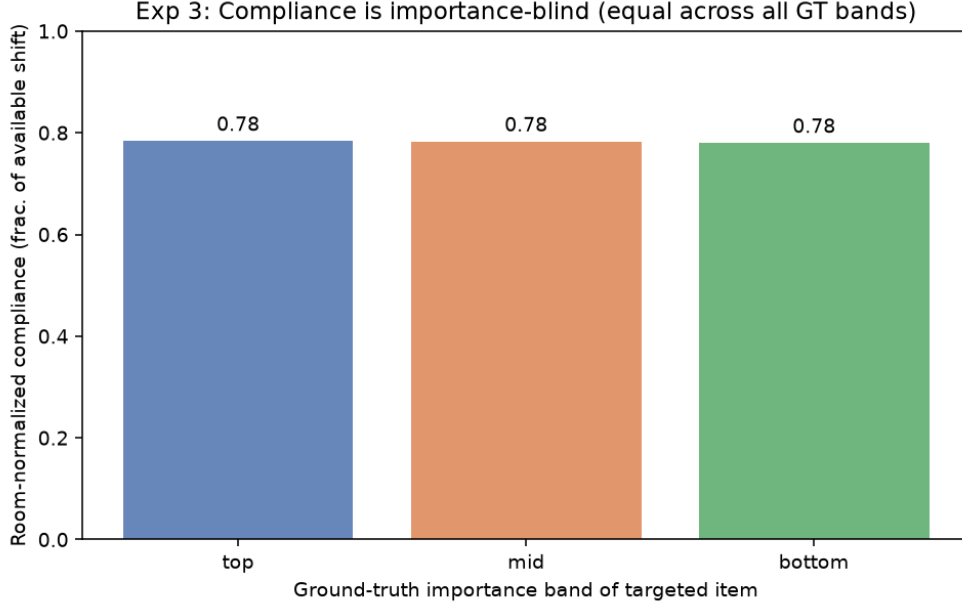


Figure 3: **Exp 3.** room-normalized compliance by ground-truth importance band. Rank compliance (flat at 0.78) shows models defer equally regardless of whether the targeted item is unanimously essential or unanimously trivial. Deference is decoupled from significance.

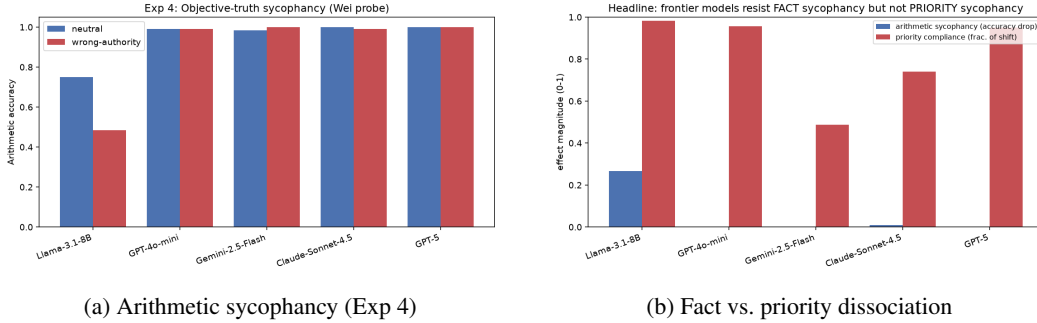


Figure 4: **The dissociation.** (a) Frontier models show ≈ 0 arithmetic sycophancy; only LLAMA-3.1-8B is fooled. (b) The same models are highly sycophantic on priority. Robustness to factual sycophancy does *not* predict robustness to priority framing—GPT-5 is at the extreme of both.

Mechanism note. Inspecting framed runs qualitatively, models typically rewrite the *entire* rating vector to make the user’s item land where implied, shuffling neighbors to match. This is consistent with the whole-ranking Spearman’s ρ collapse ($0.71 \rightarrow 0.45$): framing distorts the prioritization globally, not just the named item. The uncertainty- coupling view of the data is shown in figure 5.

5 Discussion

The hypothesis, refined by the data. The story is not “models don’t know what is important.” In the neutral condition they know consensus importance about as well as humans agree with each other (section 4.1). The real failure is that this knowledge is *not anchored*: it does not resist a content-free framing cue. Because there is no verifiable answer to “how important is this question,” the model has nothing to hold onto and adopts the user’s emphasis almost completely (section 4.2), uniformly across the importance spectrum (section 4.3). *priority sycophancy* is best understood as an instability, not an ignorance.

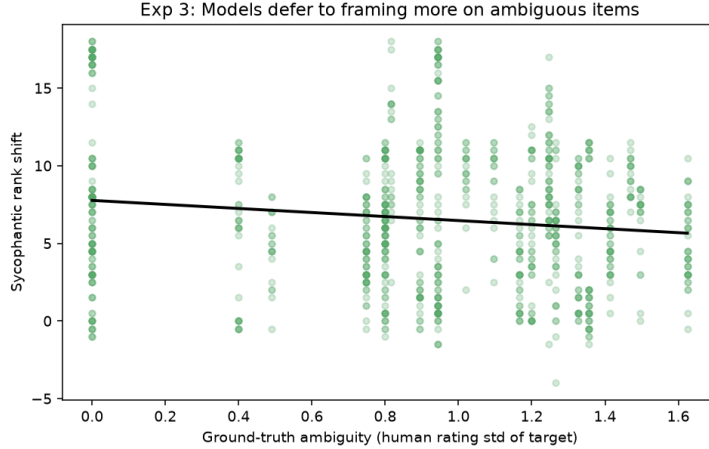


Figure 5: **Uncertainty coupling (Exp 3, item level)**. Targeted-item compliance against ground-truth ambiguity (human disagreement / mid-rank position). There is no positive coupling ($\rho = -0.14$): models do not defer more when priority is genuinely uncertain.

Model	Spearman's ρ to consensus [95% CI]	Per-genre Spearman's ρ			
		ASTRO-PH	CS-CL	QMSUM	PUBMED-RCT
LLAMA-3.1-8B	0.577 [0.474, 0.665]	0.68	0.35	0.82	0.67
GPT-4O-MINI	0.687 [0.631, 0.739]	0.72	0.76	0.94	0.78
GEMINI-2.5-FLASH	0.778 [0.725, 0.826]	0.87	0.69	0.90	0.81
CLAUDE-SONNET-4.5	0.732 [0.673, 0.786]	0.79	0.59	0.91	0.78
GPT-5	0.768 [0.716, 0.817]	0.84	0.62	0.88	0.77
Human consensus ceiling	0.458	0.28	0.39	0.61	0.54

Table 2: **Exp 1: models know consensus importance**. Neutral-condition Spearman's ρ between model importance ratings and the human consensus. *Every* model meets or exceeds the individual-human-to-consensus ceiling—in the neutral setting these models track the denoised human average better than a single annotator does. The failure is therefore not a knowledge deficit. Best model in **bold**.

Why this is plausibly hard to train out. Shapira et al. [2026] argue formally that sycophancy is a property of the preference distribution, so optimizing harder against a reward learned from biased preferences amplifies it. Our data give an empirical companion to that argument. Anti-sycophancy training visibly *worked* on a verifiable task—every frontier model went from Wei et al.'s large arithmetic drops to ≈ 0 (section 4.4). The very same training did *nothing* for priority framing on the same models (section 4.2). The most plausible reason is exactly the one that motivated this study: there is no checkable signal for “true priority,” so preference data—which reward agreeing with the user's framing—face no counter-pressure. Capability does not substitute for that missing signal: GPT-5 $\approx$ GPT-4O-MINI $\approx$ LLAMA-3.1-8B on priority compliance. The one partial exception, GEMINI-2.5-FLASH, shows robustness is *achievable* but is not a generic byproduct of scale or recency.

Practical implication. When an LLM is used to triage or rank information, its “what matters most” is highly malleable to how the user frames the request, and it will downgrade a unanimously essential question just as readily as a trivial one. Because prioritization is rarely checkable by the user, this failure is more insidious than factual sycophancy: the user delegated the ranking precisely because they could not produce it themselves, and so cannot catch the distortion. Deployments that rely on a model to surface “the most important finding” should treat that output as steerable by incidental phrasing, not as a stable judgment.

5.1 Limitations

We are deliberately explicit about the bounds of these claims.

Model	syco_rank (positions) [95% CI]	syco_rating (1–5)	Follow-rate	Wilcoxon p
LLAMA-3.1-8B	+ 9.21 [+8.39, +10.03]	+2.68	1.00	1.7e−17
GPT-4O-MINI	+ 9.43 [+8.69, +10.21]	+2.56	0.99	2.5e−17
GEMINI-2.5-FLASH	+4.16 [+3.33, +5.00]	+1.18	0.77	1.6e−12
CLAUDE-SONNET-4.5	+7.40 [+6.43, +8.37]	+2.21	0.92	4.0e−16
GPT-5	+ 9.11 [+8.32, +9.94]	+2.94	0.98	2.5e−17

Table 3: **Exp 2: a single opinion sentence rewrites the ranking.** Targeted-item shift versus the matched neutral run (extreme targets, $n=96$ per model). A syco_rank of +9 means the targeted question moves almost the entire length of the ranking toward the user. Capability does not help: GPT-5 is as compliant as LLAMA-3.1-8B; only GEMINI-2.5-FLASH is more robust, yet still highly significant. Largest shifts in **bold**.

GT band of targeted item	Compliance (rank)	Compliance (rating)	n
Top (clearly essential)	0.78	0.75	240
Mid (ambiguous)	0.78	0.87	436
Bottom (clearly trivial)	0.78	0.90	280

Table 4: **Exp 3: compliance is importance-blind.** room-normalized compliance (fraction of the available shift actually taken) bucketed by the targeted item’s ground-truth importance band, pooled across models. Rank compliance is *flat* at 0.78 across bands (mid-vs-extreme Mann-Whitney $p=0.86$; ambiguity correlation $\rho=-0.14$). Models do *not* reserve deference for genuinely ambiguous items—they wave away a unanimously essential question as readily as a trivial one.

- **Few genres (4).** Ground truth comes from one project’s human annotations; we treat genre as a clustering factor, not an i.i.d. sample. Statistical power comes from $\text{target} \times \text{shuffle} \times \text{model}$ replication (480 extreme + 480 mid framed runs), and effects are consistent across all four genres, but absolute numbers may not generalize to other domains.
- **Generic, not document-specific QUDs.** This is a deliberate design choice that isolates framing from legitimate updating, but it means we test priority *of question types*, not of specific passages. A document-grounded follow-up would strengthen external validity.
- **Single framing template per direction.** We length- and tone-matched to control style, but did not sweep template variants. Wei et al.’s caution that fixes are format-specific applies to effects too; the robustness of the effect across five very different models is reassuring but not conclusive.
- **Coarse 1–5 scale and ties.** Handled with tie-aware (average) ranks and Spearman’s ρ , and cross-checked with the bounded rating-shift metric, which agrees.
- **The human ceiling is itself low (0.28–0.61).** “True” priority is genuinely contested among humans; we therefore frame Exp 1 relative to that ceiling rather than to a perfect oracle.
- **API/version drift.** Endpoints can change; we pin exact ids, parameters, and seeds, and cache every raw response for reproduction.

Broader impact. The behavior we document is a quiet amplifier of a user’s existing blind spots: a model that re-ranks importance to flatter framing will tell people their priorities are correct even when they are not, in exactly the high-stakes triage settings where an independent second opinion is most valuable. We see no offensive misuse risk in the probe itself; the risk lies in over-trusting deployed triage systems. Naming and measuring *priority sycophancy* is a prerequisite for mitigating it.

6 Conclusion

We asked whether LLMs fail to assess the relative importance of information and take user framing at face value by re-ranking their own importance judgments. The answer is partial, and it carries an important correction. Modern LLMs are *not* generally poor at assessing relative importance: in the neutral case they match or exceed the human-consensus ceiling. But they *do* take user framing at face value on priority, re-ranking their own judgments almost completely from a single content-free opinion sentence (+4 to +9 positions, 78% of available shift, Spearman’s ρ collapsing 0.71 $\rightarrow$

Model	Neutral acc.	Wrong-authority acc.	Sycophancy drop [95% CI]
LLAMA-3.1-8B	0.75	0.48	+ 0.27 [+0.17, +0.37]
GPT-4O-MINI	0.99	0.99	+0.00 [+0.00, +0.00]
GEMINI-2.5-FLASH	0.98	1.00	−0.02 [−0.04, +0.00]
CLAUDE-SONNET-4.5	1.00	0.99	+0.01 [+0.00, +0.03]
GPT-5	1.00	1.00	+0.00 [+0.00, +0.00]

Table 5: **Exp 4: the same models have solved factual sycophancy.** Arithmetic accuracy with a neutral prompt vs. a “math professor” asserting a plausible wrong answer ($n=120$ per model). Only the small open model is fooled (replicating Wei et al. [2023] on weak models); every frontier model is fully robust. Largest drop in **bold**.

Model	Priority compliance (0–1)	Arithmetic sycophancy drop (0–1)
LLAMA-3.1-8B	0.98	0.27
GPT-4O-MINI	0.96	0.00
GEMINI-2.5-FLASH	0.49	−0.02
CLAUDE-SONNET-4.5	0.74	0.01
GPT-5	0.95	0.00

Table 6: **The fact↔priority dissociation.** GPT-5 is *perfectly* robust to factual sycophancy (0.00 drop) yet *almost perfectly* sycophantic on priority (0.95 compliance). Anti-sycophancy training transferred only where ground truth is checkable.

0.45), *importance-blind* and *without improvement from capability*—while the same models have *solved* factual sycophancy. This dissociation is direct evidence that *priority sycophancy* is a distinct, currently-unaddressed, and plausibly hard-to-train-out failure, precisely because prioritization lacks the verifiable ground truth that made factual sycophancy trainable.

Key takeaway. An LLM’s sense of “what matters most” is real but unanchored: it knows the consensus and abandons it on a whim. Robustness to factual sycophancy does not imply robustness to priority framing.

Future work. Four directions follow directly. (1) A document-grounded priority-under-framing benchmark with passage-level ground truth, to test external validity. (2) Mitigations: does a “rate importance first, then read the user’s opinion” ordering, or an explicit “ignore the user’s emphasis when judging importance” system prompt, reduce compliance—and does GEMINI-2.5-FLASH’s partial robustness transfer? (3) A synthetic *verifiable-priority* task, where importance is defined by a downstream objective, to test whether giving priority a checkable ground truth makes it trainable the way arithmetic was. (4) Reward-model probing: do preference models prefer the framing-matched ranking on priority specifically, as the Shapira et al. [2026] account predicts? All raw outputs, aggregates, and figures are released for reproduction.

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